

Simple C programs

1. **Function Overloading** (Note: C doesn't support true function overloading like C++. But we can mimic it.)
 2. **Function Overriding** (Not supported in C as it's not object-oriented — I'll show a workaround.)
 3. **Recursion** (Fully supported in C.)
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✓ 1. Function Overloading (Mimicked in C)

C doesn't support function overloading directly, but you can **simulate** it using different function names:

```
#include <stdio.h>

// Mimicking overloaded functions by using different names
int add_int(int a, int b) {
    return a + b;
}

float add_float(float a, float b) {
    return a + b;
}

int main() {
    printf("Sum of integers: %d\n", add_int(2, 3));
    printf("Sum of floats: %.2f\n", add_float(2.5, 3.7));
    return 0;
}
```

✓ 2. Function Overriding (Simulated using Function Pointers)

In C, we can simulate overriding using **function pointers**:

```
#include <stdio.h>

// Base function
void printMessage() {
    printf("This is the base function.\n");
}

// "Overridden" function
void printMessageOverride() {
    printf("This is the overridden function.\n");
}

int main() {
    // Function pointer
    void (*funcPtr)();

    funcPtr = printMessage;
}
```

```
funcPtr(); // Calls base

funcPtr = printMessageOverride;
funcPtr(); // Calls "overridden"

return 0;
}
```

✔ 3. Recursion (Example: Factorial)

```
#include <stdio.h>

int factorial(int n) {
    if (n == 0)
        return 1;
    else
        return n * factorial(n - 1); // Recursive call
}

int main() {
    int num = 5;
    printf("Factorial of %d is %d\n", num, factorial(num));
    return 0;
}
```



✔ C Language – 20 Simple Theory Interview Questions

◆ *Topic Focus: Overloading, Overriding, Recursion*

◆ Overloading (Simulated in C)

1. What is function overloading?
 2. Is function overloading possible in C?
 3. How can you simulate function overloading in C?
 4. Why does C++ support function overloading but C does not?
 5. What is name mangling and how is it related to function overloading?
 6. Can two functions in C have the same name with different parameters?
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◆ Overriding (Simulated using Function Pointers)

7. What is function overriding?
8. Why is function overriding not supported in C?

9. How can function overriding be simulated in C?
 10. What is a function pointer in C?
 11. What are the uses of function pointers?
 12. How can you call a function using a function pointer?
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◆ Recursion

13. What is recursion?
14. What is a base condition in recursion?
15. Why is a base case important in recursion?
16. What happens in memory when recursion is used?
17. What are the advantages of recursion?
18. What are the disadvantages of recursion?
19. What is the difference between recursion and iteration?
20. Can recursion be replaced by loops in all cases?